Odyssey

National cyber threat early warning

See the attack coming.

Odyssey is an AI early-warning system that tells each Indian organisation which of its systems are at risk, how sure we are, what to do and by when.

Mentor round 2: what changed since round 1, what it does on real traffic, and what we measured.

Flaws arrive faster than anyone can read

16,730

new security flaws (CVEs) were published in October 2026 alone. Every one needs a decision.

Sources: CVE Program records · NVD · CISA KEV catalogue, 2026

72% → 21%

share of new flaws that the US NVD labels with the affected product and version, January against now. Most new flaws do not say what they break.

29%

of the flaws exploited in 2026 were attacked on or before the day they were announced. Attackers move first.

One unpatched server took 300 banks offline. July 2024: ransomware entered through an unpatched Jenkins server (CVE-2024-23897, public since January, exploit code within days) at a supplier of C-Edge Technologies. About 300 small Indian banks lost UPI and ATM services. CISA listed the flaw 19 days after the banks went down. The warning signs were public for six months.

From public data to a deadline

1 Collect

CISA KEV and SSVC, NVD and vendor records, EPSS, Feodo C2, OpenPhish, DShield, CISA ICS, ATT&CK, SACHET: 13 sources, 10 real, all labelled.

2 Check

Is this exact product and version affected? NVD, the vendor's record and the advisory text are cross-checked. Advisories with no CVE id are read and matched too.

3 Score

A 0 to 100 risk with its working shown, a separate confidence score (Admiralty source grades, agreement between sources), and an SSVC tier: Act, Attend or Track.

4 Act

The CERT-In deadline set, ATT&CK mitigations, a fact-checked brief in the operator's language, STIX and CERT-In drafts, and notices a person approves.

Risk = 100 × Relevance × Importance × (55% Likelihood + 30% Severity + 15% Evidence)

The AI never sets the score.
Fixed rules do. The model writes the brief; every number, version, deadline and authority in it is checked against the facts, and a sentence that fails is withheld, never shown.

Fully offline, chosen by measurement.
One laptop, RTX 4070 with 8 GB, 0 cloud calls. Each job's model is set in a config file: MiST-8B for briefs, SecureBERT 2.0 for entities, classic ML for CWE and CVSS, IndicTrans2 for 22 languages.

Round 1 feedback, answered

1	**Make it proactive**	It runs continuously: detectors every 5 s, real feeds every 15 min, new CVEs taken in at publication, a day-0 exploitation forecast, pre-disclosure surge warnings, autopilot notices. 286 automatic actions so far, every one in the hash-chained audit trail.
2	**Real tests, real attacks**	An HTTP shield protects a real web app through a drop-in agent. A red team fires real SQLi, XSS, traversal, SSRF and brute-force requests over the wire: 45 of 45 detected, 45 of 45 blocked inline, 0 false positives, about 15 ms each.
3	**A SOC, not just a feed**	Cases open themselves, owners by rule, CERT-In SLA clocks, playbook checklists, MTTA and MTTR measured from analysts' actions, shift handover in one click. Detector thresholds re-derive from analysts' decisions, within bounds, audited.
4	**Organisation-level OSINT**	A threat map per operator: attack pressure on its ports, scanning of its products, C2 in its traffic, look-alike domains, suppliers and blast radius, every line with its source. Brand and social watch; a consented attack-surface scan.
5	**Human in the loop**	Graduated autonomy. Evidence-based notices, digests and hunts run on their own; predictions, sensor-only warnings, regulator escalations and new-advisory notices are proposed; a person approves or rejects, and the reason feeds calibration.

It fights back on real traffic

45/45

attacks detected by the HTTP shield: SQLi, XSS, path traversal, command injection, SSTI, SSRF, Log4Shell, scanning, brute force

45/45

blocked inline with HTTP 403 in block mode, the source block-listed, while the legitimate customer keeps browsing

0

false positives on benign control traffic; 100% precision

~15 ms

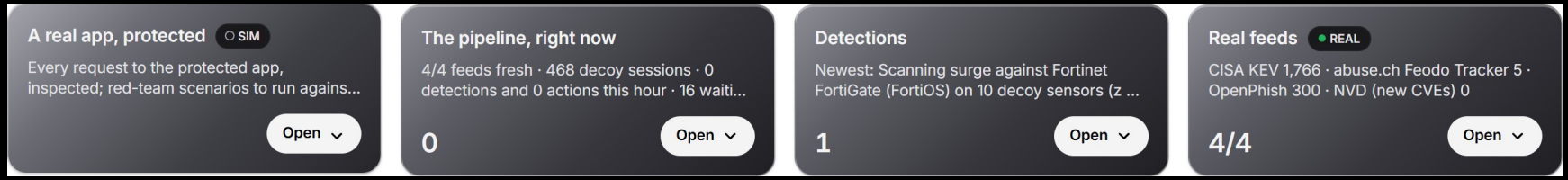
mean time to detect, each detection mapped to a CWE and an ATT&CK technique and written to the audit trail

The loop, end to end, on a target we own

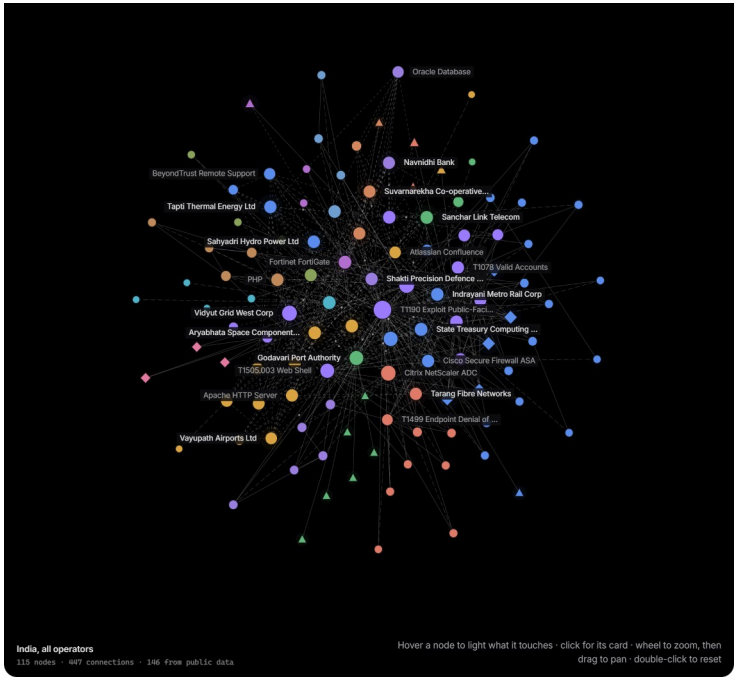
A fictional government portal (Lokseva, SIMULATION banner on every page, a mock OAuth sign-in) sits behind the shield on this laptop. A standalone red-team agent, loopback-only by code, ran three waves:

459 attacks detected · 242 blocked inline · 11 attack families, the OAuth cases included · a Compromise case opened on its own · 29 alerts delivered

Alerts leave the laptop as one audited card per event: Slack, Discord, Microsoft Teams, e-mail or any webhook. The agent refuses any target that is not loopback or private; nothing ever touches a real service.



A threat-intelligence platform, not a scanner



Intelligence graph
115 nodes, 447 connections, 146 from public data; 11 entity kinds. It breathes: fresh evidence pulses, old lies still. Replay as it stood at nine earlier instants.

Overlap, never attribution
Threat groups from the ATT&CK record, ranked by malware and technique overlap, and labelled so on every node, card and STIX object.

Watchlists and rules
Watch any entity. Fresh evidence, a worsening tier or a new connection fires the rule within 15 s and reaches the channels.

Indicators in, matched
Paste text, CSV or STIX: IPs, domains, URLs, hashes and CVEs matched in seconds against our own record, enriched from Feodo, DShield, OpenPhish and KEV.

Out to the tools you run
Risk lists as CSV or JSON, a STIX 2.1 bundle, and a read-only TAXII 2.1 server for OpenCTI, MISP or a SIEM.

Outside view, with consent
Brand and social watch from public sources (79 real hits for IRCTC). Attack-surface scan only of systems you own, graded, with fixes.

Audited against CloudSEK, Recorded Future and the field; dark-web collection, scanning of networks we do not own, takedowns and sandboxing stay out by rule.

Built for India

The clocks that apply, on every warning

- CERT-In 6-hour incident report
- RBI DAKSH
- SEBI CSCRF
- DPDP Board, 72 h
- NCIIPC
- Sector CSIRT routing

22 scheduled languages. Briefs translated on the laptop by IndicTrans2; 127 of 132 test sentences validated, and a sentence that fails is withheld, never guessed.

Supplier fan-out, the C-Edge lesson. A supplier's exposure reaches every operator it serves; the risk plan orders the fewest changes that remove the most national risk: 93 remove half, 205 remove 80%.

Weather or attack? Paired with SACHET

100
alerts this morning

99
with a state inferred

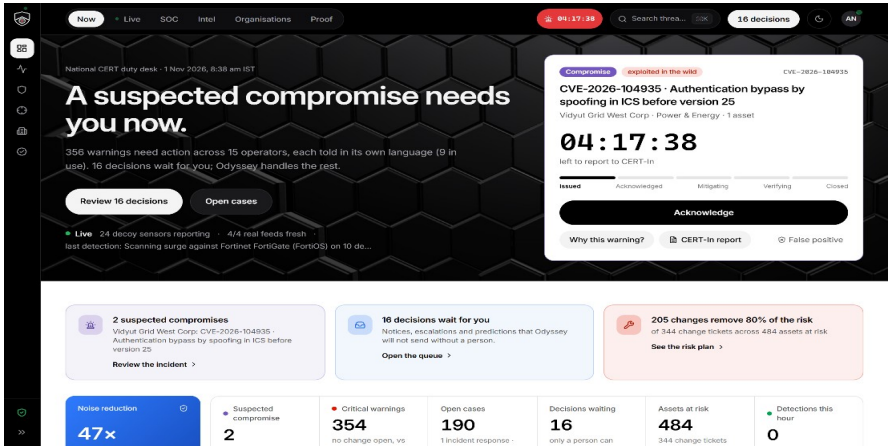
80
on our operators

NDMA's public CAP feed is read every 15 minutes, read-only and labelled REAL. Each alert's state is inferred with its basis and a confidence, and matched to the operators there, so an outage can be read next to the cyber record.

Our own warnings leave as SACHET-compatible CAP 1.2. Every inbound CAP is hostile until proven otherwise: an eight-step guard (flood, size, XXE, malformed, fields, sender, replay, expiry); 10 of 10 hostile cases refused in 0.36 s, no leak.

Not connected to SACHET; nothing is sent to NDMA. Bhashini is left out so the zero-cloud-calls claim holds.

The console, built on real feeds



Real, bundled, labelled

1,988 attacking IPs from 59 countries (DShield top 2,000, 52 M reports)

25,094 CVEs with a public exploit (Exploit-DB, 47,181 exploits)

565 Indian victims named on ransomware leak sites (ransomware.live)

1,048 breaches, 11 Indian services (Have I Been Pwned)

30 CISA ICS/OT advisories · 100 critical open-source advisories (GitHub)

3,526 criminal-network entries: Tor exits, Spamhaus DROP, ET compromised hosts

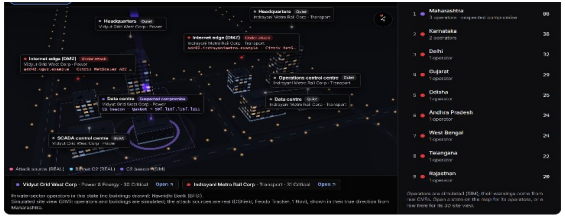
White body, black bands, in Uber's register; a night-shift mode; REAL or SIM on every panel; feeds bundled, so the demo never waits on the network.



Attack origins and the operators they reach

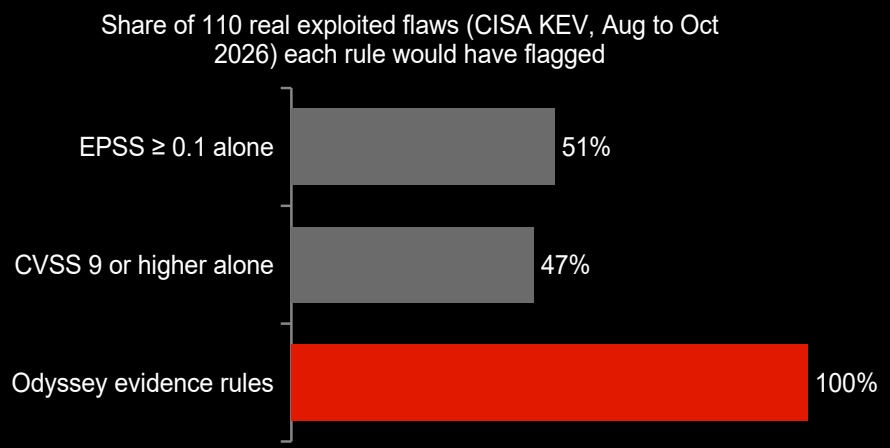


India by Critical warnings; compromise in violet



A state in 3D: its agencies and the attack origins

Measured, not claimed



Being on CISA's list is one of our rules, so 110 of 110 is by design. The honest test is the out-of-time forecast on the right: frozen on 31 July, scored on what CISA listed from August to October.

61.1%

of later-exploited CVEs caught by the day-0 forecast while flagging 2% of new CVEs; CVSS ≥ 9 at the same budget catches 16.7%

20.5 days

median lead over CISA KEV on real signals alone, up to 42.4 days

44×

less noise: 16,730 flaws in October cut to 384 Critical warnings

221

engine tests green; 26 of 26 product flows; 25 of 26 browser steps; build and lint clean

0

internet or cloud AI calls; one RTX 4070 laptop; 85 s to re-score 15 operators, 612 systems, 2,277 flaws

12 days

early warning from the decoy grid before disclosure (SIM, tuned to GreyNoise's "Ten Days Before Zero")

Honest limits. Forecast precision is 5%, because exploitation is rare (0.16% of CVEs), so forecasts are proposals for a person. Operators and sensors are simulated and labelled SIM; CVEs, advisories, attack telemetry, C2 lists and SACHET are real. State inference on SACHET alerts carries its basis and confidence and is not always right.

What is next, and what we ask of you

Next

- Real operator onboarding: inventory connectors in place of the simulated constituency
- The national supplier graph from procurement data
- Sector OT telemetry: power, water, transport
- Live CERT-In and NCIIPC feeds as they are published
- The decoy grid as a national honeynet
- Bhashini behind the translate seam, off unless credentials and consent are given

Three questions for the mentors

1. Is the open formula plus the SSVC decision path the right spine for a national system, or should sector weightings differ?
2. Where should the SACHET pairing stop: read-only context beside the cyber record, or a routed channel through an authorised agency?
3. Of the nine-minute demo (Now, Live attacks, SOC, Intel graph and indicators, SACHET, Organisations, Proof), which two minutes would you cut?

Real data, measured claims, a human in the loop, zero cloud calls, and it runs on this laptop.